A
INTERNSHIP REPORT

ON

"Ai-Learning-Assistant"

SUBMITTED TO THE SAVITRIBAI PHULE PUNE UNIVERSITY IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF THE

THIRD YEAR COMPUTER ENGINEERING

BY

Miss. Bhagwat Pooja Dilip (Seat No:.....)

UNDER THE GUIDANCE OF
Prof.Sharad.M.Rokade

DEPARTMENT OF COMPUTER ENGINEERING
Sir Visvesvaraya Institute of Technology, Nashik
A/p Chincholi, Tal. Sinnar, Dist. Nashik – 422102
Year 2025–2026

DEPARTMENT OF COMPUTER ENGINEERING

Sir Visvesvaraya Institute of Technology, Nashik

A/p Chincholi, Tal. Sinnar, Dist. Nashik – 422102

Year 2025–2026

CERTIFICATE

This is to certify that the internship report entitled

"Ai-Learning-Assistant"

is submitted as fulfillment of curriculum of the T.E. of Computer Engineering by

Miss. Bhagwat Pooja Dilip(Seat No:.....).

Prof.Sharad.M.Rokade
Internship Guide

Prof. Priyanka Y. Sanvatsarkar
Internship Coordinator

Prof. Sharad.M.Rokade
HOD

Place:SVIT, Nashik

Date:_____

Savitribai Phule Pune University

CERTIFICATE

This is to Certify that

Miss.Bhagwat Pooja Dilip (Seat No :-.....)

Student of T.E. Computer

was examined in Internship Report entitled

"Ai-Learning-Assistant"

on ..../..../2026

At

DEPARTMENT OF COMPUTER ENGINEERING,
SIR VISVESVARAYA INSTITUTE OF TECHNOLOGY,
NASHIK YEAR 2025-26

Internal Examiner External Examiner

Certificate By Guide

This is to certify that

Miss. Bhagwat Pooja Dilip (Seat No :-.....)

Has completed the report work under my guidance and that, I have verified the work for its originality in documentation, problem statement, literature survey and conclusion presented in internship report work.

Place: Nashik
(Prof.Sharad.M.Rokade)
Date:
SVIT, Nashik

INTERNSHIP CERTIFICATE

February 16, 2026

This is to certify that **Pooja Dilip Bhagwat** has successfully completed an internship at **YSM Info Solution** for the period from **January 02, 2026 to February 16, 2026**.

During the internship period, the individual was assigned to work on a project using **MERN Stack** . Throughout the tenure, strong programming skills and a self-motivated approach toward learning new technologies were demonstrated. The performance exceeded our expectations, and the individual successfully completed the learning of **project development and management practices**.

The contribution made to the organization and its success is highly appreciable. We wish the individual all the very best for future endeavors.

Regards,
For YSM Info Solution

Authorized Signatory
(Mr. Nilesh Sonawane)

YSM INFO SOLUTION NASHIK

Office No. 2, 1st floor, Dhandai Apt., Ambad Police Station Road, Nashik - 422 009.

Phone: +91-898-308-3698 | **E-mail**: info@ysminfosolution.com | **Website**: www.ysminfosolution.com

Figure 1: Internship Certificate

ABSTRACT

This internship report presents a comprehensive overview of the academic and non-academic work carried out by the intern during the internship period. The internship was undertaken at Netleap it training and solutions from 1/1/26 to 14/2/26 as part of the TE Computer Engineering curriculum. The primary objective of the internship was to gain practical exposure to real-world applications of computer technology and to apply theoretical knowledge in a professional environment. During the internship, the intern worked on the development of an Ai-Learning-Assistant for Symptom Detection, involving tasks such as data analysis, machine learning model development, Natural Language Processing, and user interface design. The internship provided hands-on experience with modern tools, technologies, and development practices, enhancing technical, analytical, and problem-solving skills.

Keywords An AI Learning Assistant can be described through a set of interconnected theoretical keywords that reflect its conceptual foundation in modern computing and education systems. At its core, it embodies Artificial Intelligence as the driving force enabling intelligent behavior, supported by Machine Learning techniques that allow the system to learn from user interactions and improve over time. The concept relies heavily on Natural Language Processing (NLP), which facilitates meaningful communication between the user and the system by interpreting queries and generating human-like responses.

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to the team at our organization for providing us with the opportunity to work on this AI Learning Assistant Web project. Their guidance, support, and valuable feedback were instrumental in the successful completion of this project.

We extend our special thanks to our project mentor, Prof. Sharad.M.Rokade, for continuous encouragement and expert guidance throughout the development of the AI Learning Web platform. I am highly indebted to Principal Dr. Sarang S. Pande for the facilities provided to accomplish this project successfully. I would like to thank my Head of the Department, Prof. Sharad M. Rokade, for his constructive suggestions and support throughout the project.

I would also like to thank Prof. Priyanka Y. Sanvatsarkar, Internship Coordinator, Department of Computer Engineering, for her support and guidance in completing this project. I am extremely grateful to my department staff members and friends who helped me in the successful completion of this project.

Miss. Bhagwat Pooja Dilip
SVIT, Nashik

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Chapter 1

SUMMARY

In January, I completed my internship at Netleap IT Training and Solutions, where I worked on the project titled "AI Learning Assistant using MERN Stack." This internship provided me with practical exposure to Full Stack Development and Artificial Intelligence integration in modern web applications.

The primary objective of the project was to design and develop an intelligent learning assistant that helps users understand concepts, answer questions, and provide personalized learning support. The system processes user queries, interacts with AI-based services, and delivers accurate and meaningful responses.

During the internship, I worked on frontend and backend development, API integration, database management, and deployment. I gained hands-on experience in MongoDB, Express.js, React.js, Node.js, and AI API integration.

The offline mode of the internship allowed direct interaction with mentors, enhancing learning and collaboration.

Overall, this internship improved my technical knowledge, problem-solving skills, and confidence in developing AI-powered web applications.

1.1 Full Stack Internship

The Full Stack internship provided practical exposure to developing modern web applications using the MERN stack integrated with AI capabilities. The project focused on building an AI Learning Assistant for educational support.

1.2 Need of the Project

The need for this project arises due to the rapid growth of the information technology industry and the increasing demand for skilled full stack developers. In today's competitive environment, theoretical knowledge alone is not sufficient; students must

have practical experience in developing real-world applications. This project helps in bridging the gap between academic learning and industry requirements by providing hands-on exposure to modern technologies and development practices. It enables students to understand how different components such as frontend, backend, and databases work together to form a complete system.

Additionally, the project is important for improving problem-solving abilities, logical thinking, and coding skills. It allows students to face real-time challenges such as debugging errors, managing data, and optimizing performance. Through this project, students also learn how to use development tools, frameworks, and version control systems, which are essential in professional environments. The experience gained helps in building confidence and prepares students for job interviews and technical assessments. Moreover, with the integration of advanced technologies like Artificial Intelligence, the project becomes more relevant in today's digital world. The AI-based learning assistant helps in enhancing user experience by providing intelligent and automated solutions. This highlights the importance of combining web development with emerging technologies to create innovative applications.

1.3 New Technology

The project makes use of modern and advanced technologies that are widely used in the current software development industry. One of the key technologies used is the MERN Stack, which consists of MongoDB, Express.js, React.js, and Node.js. This stack is highly popular for building dynamic and scalable web applications. MongoDB is a NoSQL database that stores data in a flexible, JSON-like format, making it efficient for handling large amounts of data. Express.js and Node.js are used for backend development, where Node.js provides a runtime environment to execute JavaScript on the server side, and Express.js simplifies the process of building APIs and handling server requests. React.js is used for frontend development and helps in creating fast, responsive, and interactive user interfaces.

Another important technology used in the project is Artificial Intelligence. AI enhances the functionality of the application by enabling features such as automated responses, smart recommendations, and intelligent query handling. In the AI Learning Assistant system, AI helps users by providing instant answers to their questions and suggesting relevant study materials. This makes the application more interactive and user-friendly.

The project also utilizes cloud-based technologies for deployment and hosting. Platforms such as Netlify, Vercel, or Render are used to deploy the application, making it accessible from anywhere. Cloud technology ensures scalability, reliability, and easy maintenance of the system. Additionally, version control tools like Git and GitHub

are used to manage code efficiently and support collaboration among developers.

Overall, the use of these new technologies makes the project more efficient, scalable, and aligned with current industry standards. It also helps students gain knowledge of modern tools and frameworks, which is essential for building innovative applications and succeeding in the IT field.

1.4 Drawbacks

Although the project offers many advantages and learning opportunities, it also has certain drawbacks and limitations that need to be considered. One of the major challenges is the complexity of full stack development, as it requires knowledge of multiple technologies such as frontend frameworks, backend programming, and database management. For beginners, understanding and managing all these components simultaneously can be difficult and time-consuming. Additionally, debugging errors in a full stack application can be complicated because issues may arise at different levels, including the user interface, server, or database, making it harder to identify and fix problems efficiently.

Another drawback is the dependency on internet connectivity and modern tools. Since the project involves cloud deployment, API integrations, and version control systems, a stable internet connection is necessary for smooth development and testing. Any interruption in connectivity can slow down the workflow. Moreover, performance issues may occur if the application is not properly optimized, especially when handling large amounts of data or multiple users simultaneously.

Security is also a significant concern in full stack applications. If proper measures such as authentication, data encryption, and validation are not implemented correctly, the system may become vulnerable to cyber threats like data breaches or unauthorized access. Ensuring security requires additional effort and knowledge, which can be challenging for students during the learning phase.

Chapter 2

INTRODUCTION

2.1 Basic of Internship

The internship provides a strong foundation for understanding the practical aspects of full stack development. It focuses on hands-on learning, where students actively participate in designing and developing web applications instead of only studying theoretical concepts. During the internship, students are introduced to various tools, technologies, and frameworks that are used in real-world software development. They learn how to write clean and efficient code, debug errors, and optimize application performance.

The internship also helps students understand the workflow of a software development process. It includes requirement analysis, designing the system, coding, testing, and deployment. Students gain experience in using development tools such as code editors, version control systems like Git, and platforms like GitHub for collaboration. Additionally, the internship encourages teamwork, communication, and time management skills, which are essential in a professional environment. Overall, it serves as a bridge between academic learning and industry practices.

2.2 Objective of Internship

The main objective of the internship is to provide practical exposure to full stack development and help students gain industry-relevant skills. It aims to enhance the technical knowledge of students by allowing them to work on real-time projects and modern technologies. The internship focuses on developing the ability to build complete web applications by integrating frontend, backend, and database systems.

Another important objective is to improve problem-solving skills and logical thinking. Students learn how to analyze problems, design solutions, and implement them effectively. The internship also aims to familiarize students with the software devel-

opment life cycle, including planning, designing, coding, testing, and maintenance. Furthermore, it helps students build confidence, improve communication skills, and prepare for job placements. By the end of the internship, students are expected to have a clear understanding of real-world development practices and be ready for professional roles.

2.3 Internship Topic

2.3.1 AI Learning Assistant using MERN Stack

The selected internship topic focuses on developing a modern web-based application using advanced technologies. The topic is chosen to align with current industry trends and to provide maximum learning opportunities for students. It involves the use of full stack development concepts along with emerging technologies to create a useful and innovative system.

The project aims to solve real-world problems by providing an efficient and user-friendly solution. It includes various functionalities such as user interaction, data management, and system processing. The topic is designed in such a way that it covers all major aspects of software development, including frontend design, backend logic, and database integration. This ensures that students gain a complete understanding of how applications are built and managed.

Chapter 3

OVERVIEW OF COMPANY

3.1 History of Company

The company where the internship is conducted has been established with the aim of providing quality IT services and technical training to students and professionals. Initially, the company started as a small organization focusing on basic software development and training programs. Over time, it expanded its operations and began offering advanced services in various domains such as web development, machine learning, and database management.

With continuous growth and innovation, the company has built a strong reputation in the field of technology and education. It has successfully trained many students and helped them gain practical knowledge and industry exposure. The company focuses on bridging the gap between academic learning and industry requirements by providing real-time project experience and mentorship.

3.2 Products and Services

3.2.1 Internship Programs

The internship programs offered by the company are designed to provide hands-on experience in various technologies such as full stack development, artificial intelligence, and data science. These programs focus on practical learning through real-time projects, allowing students to apply their theoretical knowledge in real-world scenarios. The internships also include mentorship, guidance, and evaluation to ensure that students gain maximum benefit from the training.

Students are provided with opportunities to work on live projects, collaborate with peers, and learn industry-standard practices. At the end of the internship, they receive certification, which adds value to their resume and helps in career growth.

3.2.2 Machine Learning Solutions

The company provides machine learning solutions that help in building intelligent systems capable of analyzing data and making predictions. These solutions are used in various applications such as recommendation systems, chatbots, and predictive analytics. The company uses advanced tools and technologies to develop efficient and accurate models.

3.2.3 Database Management

Database management is an essential service provided by the company to ensure efficient storage, retrieval, and security of data. The company designs and manages databases using both SQL and NoSQL technologies, depending on the requirements of the application. Proper database management helps in maintaining data integrity, reducing redundancy, and improving system performance.

The company also focuses on data security by implementing measures such as authentication, authorization, and backup systems. This ensures that sensitive information is protected and easily recoverable in case of system failures. Database management plays a crucial role in the overall performance and reliability of any application.

3.2.4 Cloud Computing

Provides training in cloud technologies and deployment.

Chapter 4

INTERNSHIP ACTIVITY

4.1 Technologies Learned

During the internship, various technologies were learned that are essential for full stack development. These include frontend technologies such as HTML, CSS, JavaScript, and React.js, which are used to design and develop user interfaces. Backend technologies like Node.js and Express.js were used to handle server-side operations and create APIs. MongoDB was used as a database for storing and managing application data. In addition to these, tools like Git and GitHub were used for version control and collaboration. Students also gained exposure to deployment platforms such as Netlify and Vercel. Learning these technologies helped in understanding how different components of a web application work together and provided practical experience in building real-world projects.

4.2 Software Requirements

The successful development of the project required various software tools and environments. A code editor such as Visual Studio Code was used for writing and editing code. A web browser like Google Chrome was used for testing and debugging the application. Node.js was installed to run backend applications, and MongoDB was used for database management.

Other tools included Git for version control and platforms like GitHub for storing and sharing code. Additionally, deployment platforms such as Netlify or Render were used to host the application online. These software requirements ensured smooth development, testing, and deployment of the project.

4.3 Daily Activities

The daily activities during the internship involved a combination of learning, coding, and project development. Each day typically started with understanding new concepts or reviewing previously learned topics. Students then worked on coding tasks related to frontend or backend development. This included designing user interfaces, writing API logic, connecting databases, and testing application functionality.

Debugging errors and improving code quality were also important parts of daily work. Regular interaction with mentors helped in resolving doubts and gaining better understanding of concepts. Students also updated their progress using version control systems like Git. These daily activities helped in maintaining consistency and ensured steady progress in the project.

Chapter 5

ASSESSMENT

5.1 Skills Gained

The internship helped in developing both technical and non-technical skills, which are essential for a successful career in the IT industry.

5.1.1 Technical Skills

Technical skills gained during the internship include proficiency in programming languages such as JavaScript and understanding of web development frameworks like React.js and Node.js. Students learned how to build responsive user interfaces, create APIs, and manage databases using MongoDB. They also gained knowledge of version control systems like Git and deployment techniques.

Additionally, students developed problem-solving skills by working on real-time projects and debugging errors. They learned how to optimize application performance and ensure security in web applications. These technical skills are highly valuable and relevant in the software development field.

5.1.2 Soft Skills

Apart from technical skills, the internship also helped in improving soft skills. Communication skills were enhanced through interaction with mentors and team members. Teamwork and collaboration were developed while working on group tasks and projects.

Time management skills were improved by completing tasks within deadlines. Students also learned to adapt to new challenges and work in a professional environment. These soft skills are essential for personal and professional growth.

5.2 Responsibilities

During the internship, students were assigned various responsibilities related to project development. These included designing the front end interface, developing backend API s, and managing the database. Students were also responsible for testing the application and fixing errors to ensure smooth functionality.

They had to follow coding standards and maintain proper documentation of their work. Regular updates and progress reports were shared with mentors. These responsibilities helped students understand the importance of discipline, accountability, and teamwork in a professional environment.

Chapter 6

CONCLUSION

The Full Stack Internship provided a valuable learning experience by combining theoretical knowledge with practical implementation. It helped in developing a strong understanding of web development technologies and the complete software development process. Through hands-on experience, students were able to build real-world applications and improve their technical and problem-solving skills.

The internship also contributed to personal growth by enhancing communication, teamwork, and time management skills. Overall, it played an important role in preparing students for future career opportunities in the IT industry and building confidence to handle real-world challenges.

Chapter 7

FUTURE WORK

The project can be further improved by adding more advanced features and functionalities. Future enhancements may include improving the user interface for better user experience, adding more AI-based features for accurate and intelligent responses, and integrating advanced technologies such as machine learning models.

The system can also be made more scalable to handle a larger number of users. Security features can be enhanced to protect user data more effectively. Additionally, mobile application development can be considered to make the system accessible on smartphones. These improvements will make the application more efficient and user-friendly.

Chapter 8

PROJECT PERFORMED

8.1 AI Learning Assistant

The AI Learning Assistant is the main project developed during the internship. It is a web-based application designed to help students in their learning process by providing instant answers and study support. The system uses modern web technologies and AI features to create an interactive and efficient learning platform.

8.1.1 Features

The application includes several important features such as user registration and login, interactive user interface, AI-based question answering system, and access to study materials. It provides quick and accurate responses to user queries and helps students understand concepts more effectively.

8.1.2 Development Steps

The development of the project followed a structured approach. It started with planning and requirement analysis, followed by designing the user interface. Then, frontend development was carried out using React.js, and backend development was done using Node.js and Express.js. MongoDB was used to store data, and APIs were created to connect different parts of the system. Finally, the application was tested and deployed using cloud platforms.

8.1.3 Outcome

The system successfully provides intelligent learning assistance and improves user experience through AI.

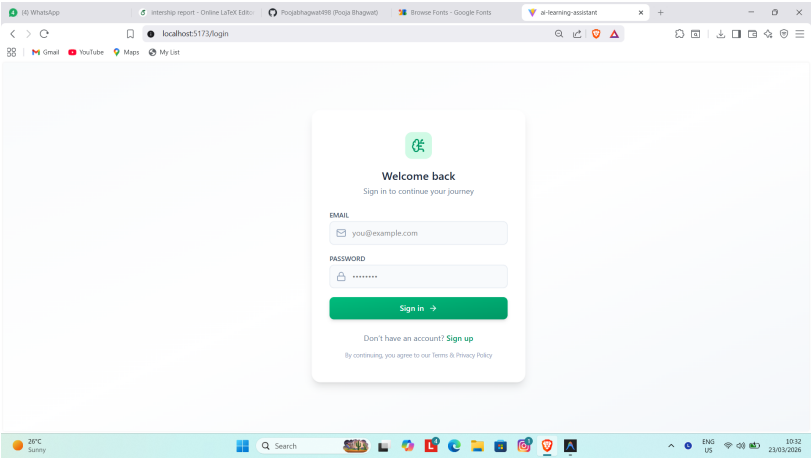

Figure 8.1: Login page

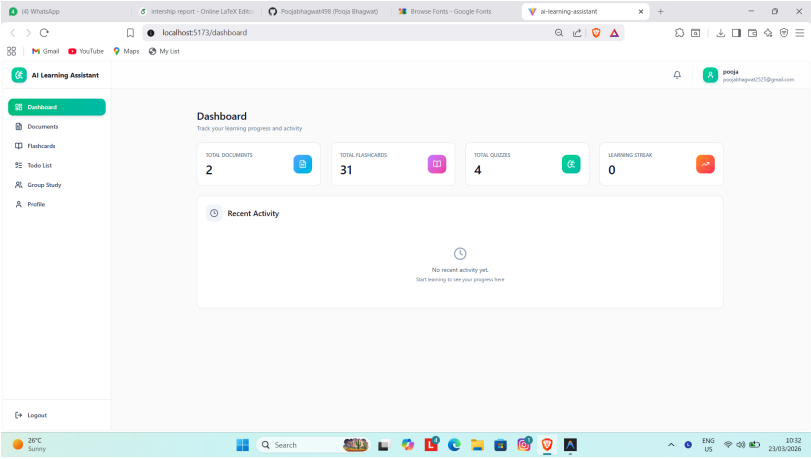

Figure 8.2: Dashboard

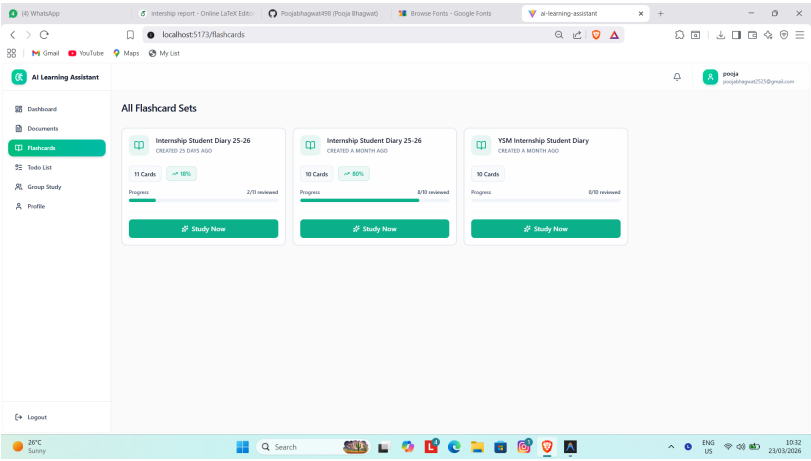

Figure 8.3: All FlashCards

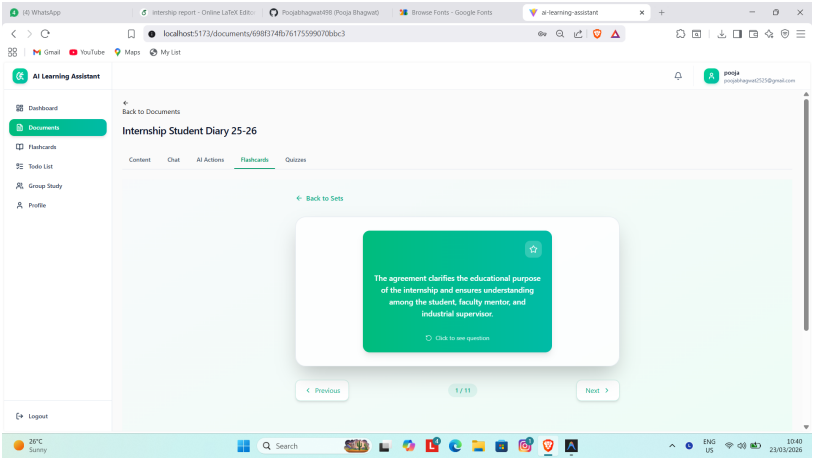

Figure 8.4: Flash Cards

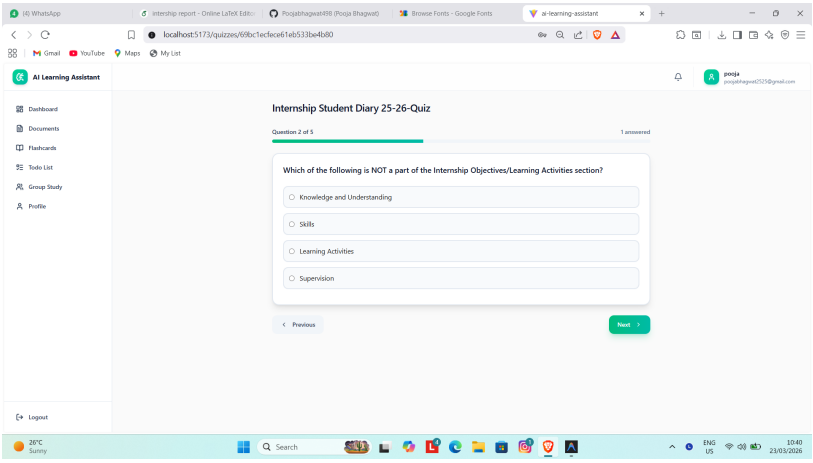

Figure 8.5: Quizzes

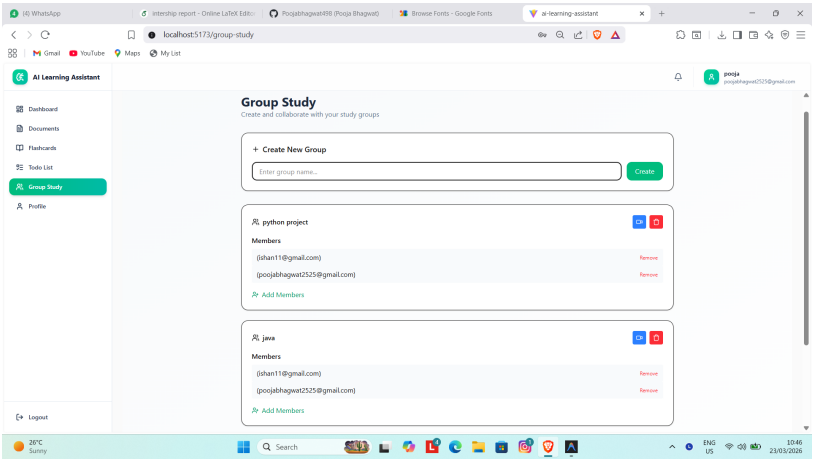

Figure 8.6: Group study